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PROFESSIONAL SUMMARY

Software Engineer with a focus on AI/ML and NLP, having built end-to-end projects including Agentic RAG systems, intelligent document assistants, and financial sentiment analysis pipelines. Proficient in Python, LangChain, LangGraph, FastAPI, and vector databases, with practical experience integrating LLMs into production-ready applications.

TECHNICAL SKILLS

Programming: Python, JavaScript, SQL

AI/ML: Machine Learning, NLP, RAG Pipelines

Frameworks & Libraries: FastAPI, LangChain, LangGraph, RAGAS, Scikit-learn, HuggingFace Transformers, Pandas, NumPy

LLMs & Vector DBs: Groq LLM, Qdrant, Embedding Models, BERT

Tools & Platforms: Docker, Git, GitHub, Postman, Kubernetes, GitHub Actions, MongoDB, SQL Server

EXPERIENCE

CODIRO Technologies

January 2026 – February 2026

MERN Stack Developer Intern

Pakistan

- Developed responsive frontend interfaces for the CEF and Luxe Balasam websites using Next.js.
- Worked on component architecture, page routing, and API integration with a focus on performance and clean code.

PROJECTS

Agentic Research Assistant with LangGraph RAG Pipeline

2026

- Built an Agentic RAG research assistant using LangGraph and LangChain with a multi-step workflow for retrieval, evaluation, and response generation.
- Designed a complete pipeline including document ingestion, text chunking, embedding generation, vector search, query routing, and LLM-based answer generation.
- Implemented intelligent routing and fallback workflows that evaluate retrieved context and trigger web search when relevant information is unavailable in the knowledge base.
- Integrated state management and modular AI nodes to support semantic search, query handling, and structured research responses.

AI-Powered RAG Document Assistant

2026

- Built a full-stack intelligent document assistant using Python (FastAPI) and LangChain that allows users to upload PDF, DOCX, CSV, and TXT files and receive AI-generated answers with citations.
- Implemented a complete Retrieval-Augmented Generation (RAG) pipeline with semantic search using Qdrant vector database and HuggingFace embeddings.
- Integrated Groq LLM with real-time streaming responses for fast and context-aware question answering.
- Designed scalable document ingestion, chunking, embedding generation, and retrieval workflows for efficient knowledge extraction.

Stock Sentiment Analysis using NLP & Machine Learning

2025

- Built an end-to-end NLP pipeline to analyze financial news headlines and predict stock market movement using TF-IDF, Logistic Regression, BERT embeddings, and XGBoost.
- Performed text preprocessing, feature engineering, sentiment analysis, and comparative model evaluation on real-world financial datasets.
- Implemented machine learning workflows for training, validation, and performance optimization of NLP models.
- Visualized insights and model performance metrics to compare traditional ML techniques with transformer-based embeddings.

EDUCATION

FAST-NUCES

BS Software Engineering

Islamabad, Pakistan

June 2025